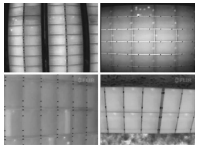
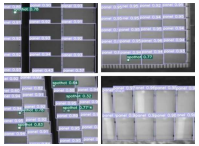
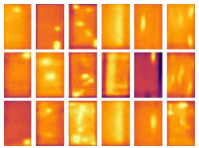
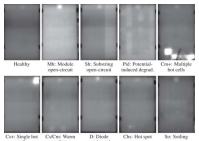
Task Description	Data				Model			
	Sample	Size	Annotations	Source	Algorithm	Architecture	Performance	Ref
Hot-Spot (HS) Detection in PV Modules		14,215 frames 640 × 480 x 1 336 × 256 x 1 320 × 240 x 1	3,654 Local HS 76 Global HS	Available Upon Request [145]	Explicit DIP	Raindrop	F1 = 77.0%	[145]
Hot-Spot (HS) and PV Modules Detection		5,600 frames 640 × 480 x 1 336 × 256 x 1 320 × 240 x 1	8,070 Local HS 125,722 PV modules	Not Available	Supervised DL	S-YOLOv5	F1 = 96.6%	[146]
PV Modules Thermal Pattern Classification		20,000 IR Patches 40 x 24 x 1	12 Classes of Anomalies	Openly Available [147]	Supervised DL Supervised DL	Ensemble CNN ResNet based Custom CNN 4 Conv2D	F1 = 74,92% F1 = 73.73%	[148] [149]
PV Modules Segmentation		862 frames 640 x 512 x 1	26,612 PV modules	Code and Data Sample Available [150], [151]	Supervised DL + DIP Supervised DL + DIP	Mask R-CNN Mask R-CNN + GPS + SfM	mAP50 = 87.8% mAP50 = 99.3%	[152] [153]
PV Modules Thermal Pattern Classification		4.16 million IR Patches 640 x 512 x 1	12 Classes of Anomalies	Available Upon Request [154]	Supervised DL Semi- -Supervised	ResNet-50 ResNet-34 + kNN Classifier	F1 = 90.68% AuROC from 73.3% to 96.6%	[152] [154]